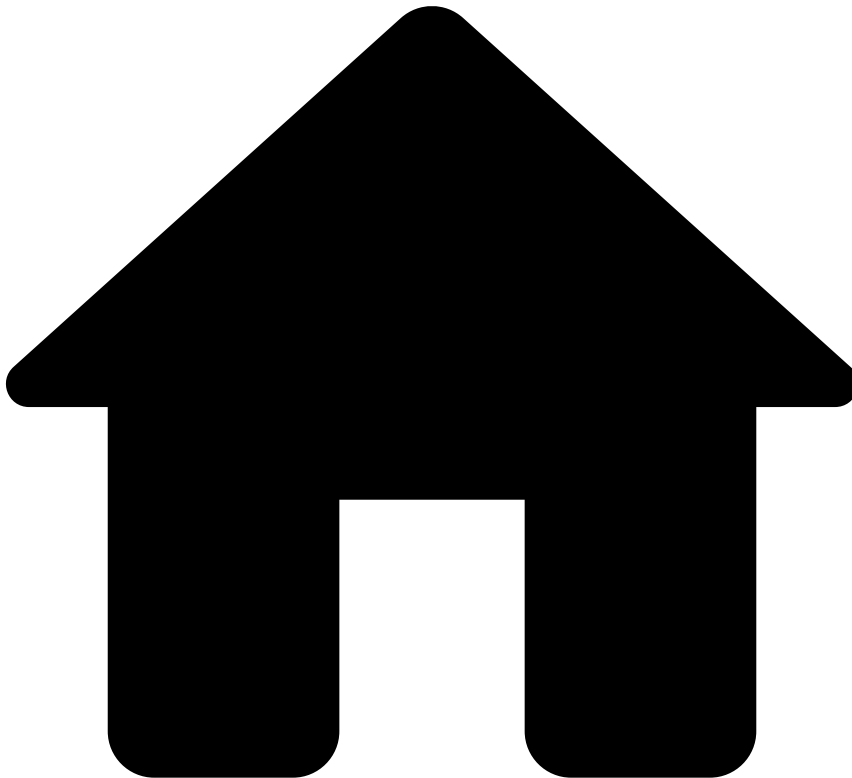


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- [Core Concepts](#)
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On this page

Operator Reference

Was this page helpful? 

When designing a [plan](#) to [process your data](#), the following operators are available:

Operators in deployable plans

Source

Defines an original data source the plan takes as input. Uses the schema of the data source to define the input format for the plan.

A [non-deployable plan](#) can take several sources, for example, coming from different files.

Mapâ

Performs major data transformations on your datasets. Lets you add new fields to the current data schema, remove existing ones, and update expressions that define how the value of a field is calculated. You can also reorder the fields on the "Add Fields" tab of a map operator in a manner that better suits your work.

Use the map to manipulate your data, [clean up your data](#), transform your data, perform calculations on existing fields to [derive features](#), etc.

You can configure field additions and updates with advanced expressions implemented in [UDFs](#).

Filterâ

Applies an expression to the data and narrows down its scope.

Use the filter to, for example, obtain only transactions from a specific merchant that require different processing.

You can apply filters with expressions implemented in [UDFs](#).

Listâ

Accesses the content of a list and stores it to a new field which gets added to the events that traverse the list operator.

Metricsâ

Generates measurable information, aggregates results and gives you valuable insight on your data, for the [time windows](#) you define.

Calculates all selected aggregations for all possible combinations of selected time windows and grouping fields.

Available aggregations are **avg**, **count**, **max**, **min**, **sum**, as well as advanced expressions implemented in [UDFs](#).

Use metrics, for example, when in a [features](#) plan you want to calculate the total amount (**sum**) spent by each card (**Group By**), in a single day (**Window**).

Sampleâ

Reduces the current data to a configurable percentage of the original size. Sampled values are randomly selected using the sampling percentage as the selection probability.

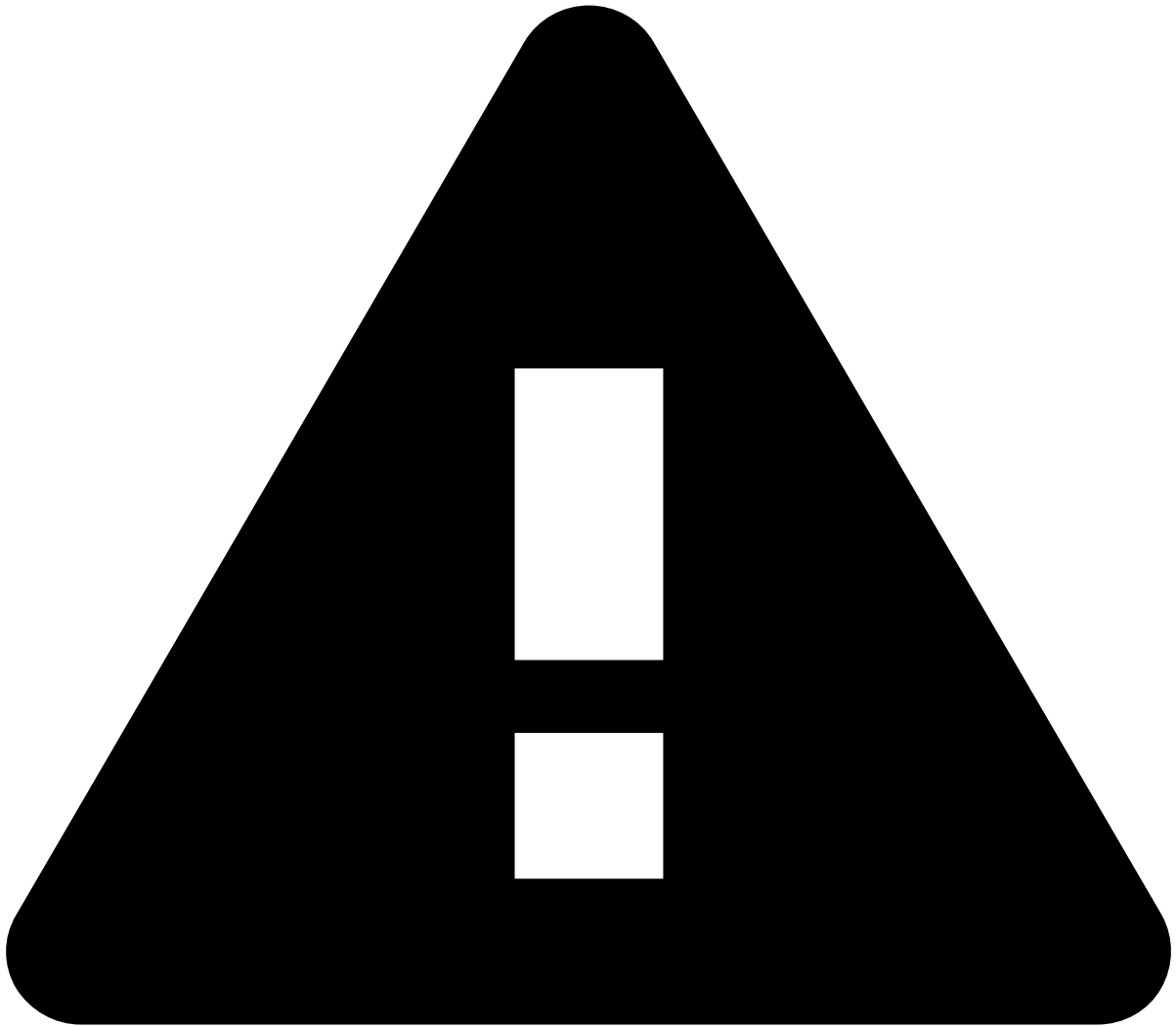
Sampling a dataset multiple times, with the same percentage, will always return the same resulting data.

Use the sample, for example, when implementing a [Stratified Sample](#) to reduce the number of non-fraud transactions in your dataset.

Sinkâ

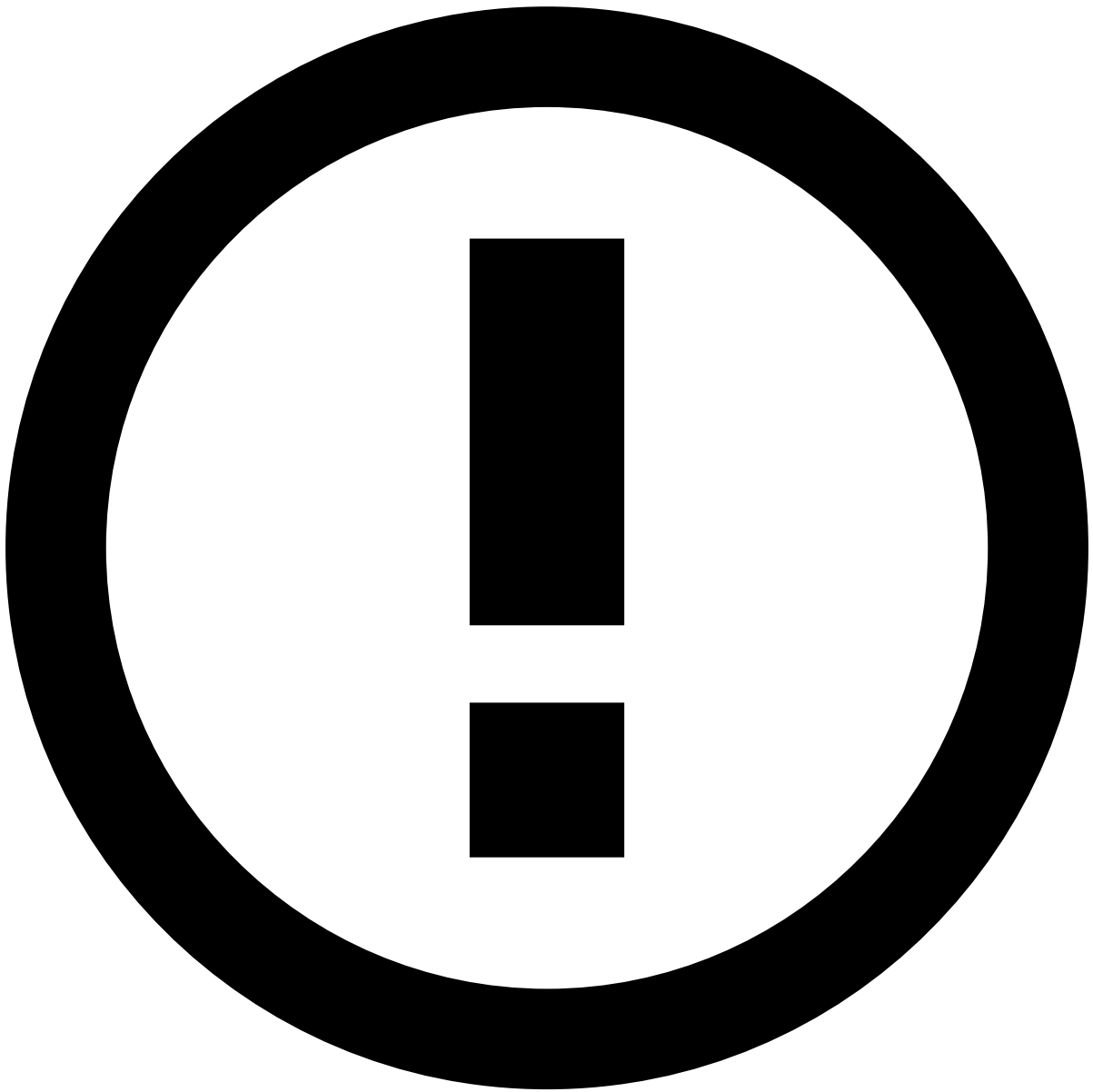
Terminates the logical flow of the plan.

When you run the plan, the sink outputs the result of the plan to a new data source and to a file in your filesystem.



Non-deterministic functions

When you apply UDFs in Map, Filter or Metrics operators, the usage of non-deterministic functions can result in non-deterministic results. For example, when you include any randomness factor into your expressions.



Source and sink operators

Pulse automatically adds a source and a sink operator to deployable plans. You need only to configure them when executing the plan.



Clipboard

You can use the clipboard in order to copy-paste operators in plans. This helps you save time by allowing you to create plans more efficiently without the need of specifying all their fields each time you create a new plan, letting you use the operators you have already configured.

Operators in non-deployable plans

Some operators only make sense at a design and exploration stage in your data science project. When processing a data stream in real-time to classify instances as `true` or `false` (let's take the fraud detection use case as an example), these activities are not repeatable nor applicable, for instance when joining data from separate sources.

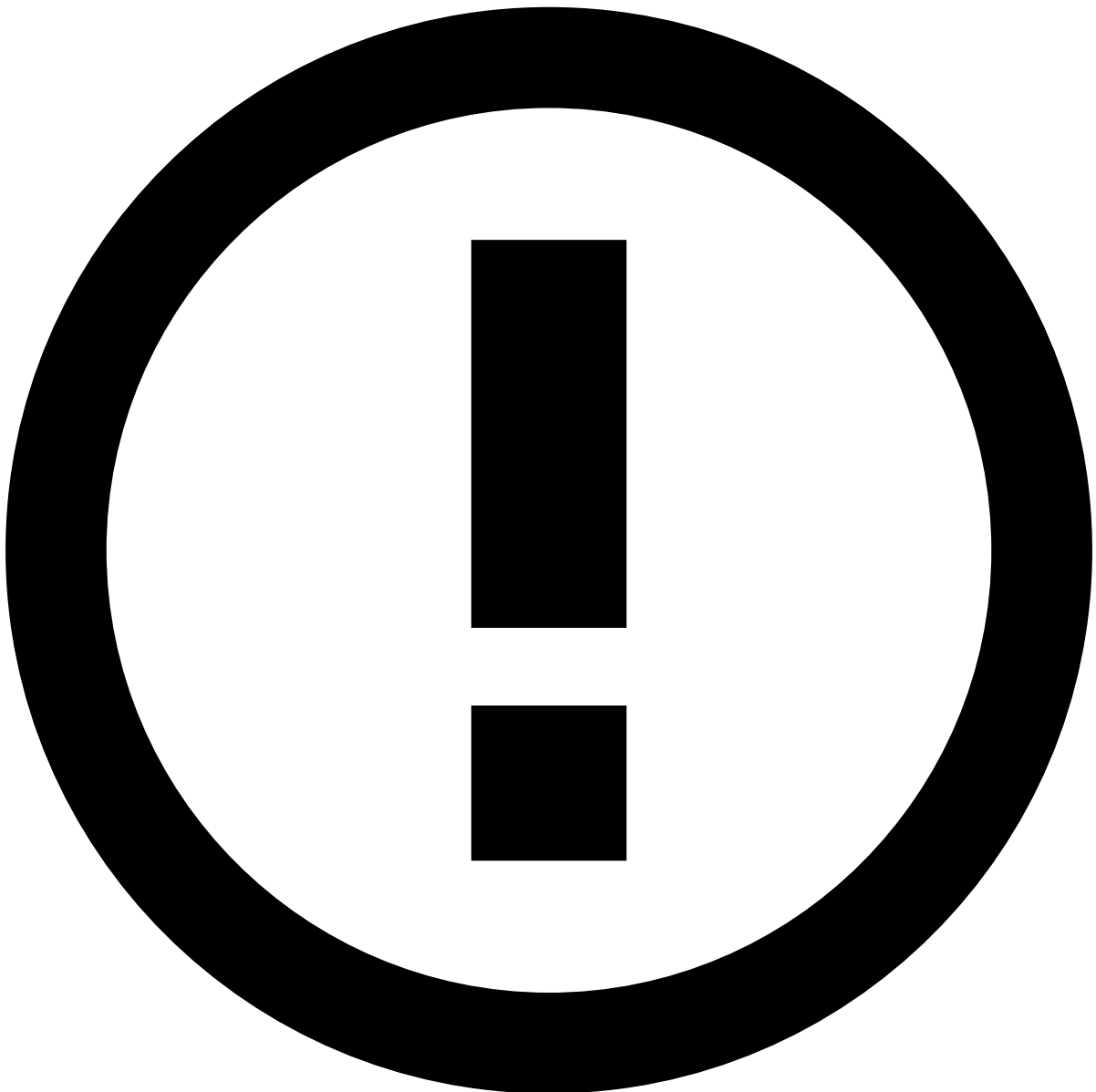
As such, the following operators are only available for [non-deployable plans](#), and perform typical data manipulation activities you might do in a database:

Group By

Groups data on a combination of fields, calculates aggregated metrics for each grouping, and outputs the result to a new field in the resulting dataset. Available aggregations are **avg**, **count**, **max**, **min**, **sum**.

For example, group by *card_number* and use the *max(amount)* aggregation over a one-month window to obtain the highest transaction amount for each card in the last month.

Lookup By



Availability of Lookup Metrics

The **Lookup By** operator is only available when using the [Railgun engine](#), and when explicitly enabled via configuration. To activate this functionality, reach out to the Feedzai Customer Success (CS) team.

Extends the capabilities of the **Group By** operator by providing a way to select which group to return the aggregated value of. (By default, **Group By** always returns the aggregation that matches the value of the grouping fields in the current row.)

The `Lookup By` operator allows specifying a retrieval field (or set of fields) that is different from the grouping field(s), thus overriding the default behavior of the `Group By` operator.

For example, group the data by *recipient_ID* and use the *sum(amount)* aggregation over a one-week window, and then use *sender_ID* as the lookup field to obtain the total amount received in the last week by the account that is sending money in a transaction.

Join

Combines rows from two workflow data paths that match on the specified fields.

Allows for **Left**, **Inner**, **Right** and **Full** join.

Union

Combines the resulting data from two workflow data paths.

Both datasets must have the exact same fields with the same types, which can be specified in a different order. The output schema can be chosen when configuring the union.